EpiSurveil: A Modular Python Framework for Epidemic Surveillance, Sequential Bayesian Estimation, and Particle-Filter-Guided Optimal Intervention Planning

Florent Ouabo Kamkumo

July 17, 2026

Abstract

Background: Linking mechanistic epidemic models to heterogeneous surveillance data streams in real time, and translating the resulting estimates into actionable intervention recommendations, requires software that integrates disease dynamics, observation models, sequential inference, and decision logic in a coherent and testable manner. Existing tools address individual layers but rarely combine all four.

Methods: We present `EpiSurveil`, an open-source Python platform that integrates (i) a nine-compartment SVEAIHCRD model with multipatch force of infection, (ii) a Bootstrap Particle Filter (BPF) with systematic resampling and tempered multi-channel likelihoods tracking five time-varying parameters jointly with the epidemic state, (iii) a reporting-delay convolution pipeline for case-channel alignment, (iv) a SARI-sentinel panel-extension methodology that extrapolates hospitalisation surveillance beyond mandatory reporting periods, and (v) a two-control optimal intervention module combining non-pharmaceutical intervention intensity u_t with testing/detection intensity v_t, solved via direct transcription (L-BFGS-B) for single-horizon planning and via Particle-Filter Model Predictive Control (PF-MPC) for adaptive receding-horizon decision support.

Results: Applied to German COVID-19 data (10 March 2020 to 7 October 2024; 1 673 days extended with RKI COVID-SARI sentinel data), the five-parameter BPF ($N = 2\,000$ particles; dynamic β_t, $\tau_{I,t}$, $\delta_{H,t}$, $\rho_{C,t}$, $Q_{C,t}$) achieves case-channel 80 % coverage of 94.9 %, ICU 70.1 %, and mean ESS = 1 259. The two-control optimiser converges in under 20 seconds for a 90-day horizon (180 variables); PF-MPC over a 60-day retrospective period with a 14-day lookahead runs in ∼17 seconds and averts 1 430 deaths relative to an uncontrolled baseline.

Conclusions: `EpiSurveil` provides a transparent, modular foundation for multi-source epidemic data assimilation and uncertainty-aware intervention planning. Its layered design allows components to be replaced independently, supporting both methodological research and operational public health applications.

Keywords: epidemic surveillance; particle filter; Bayesian data assimilation; optimal control; model predictive control; COVID-19; testing strategy; non-pharmaceutical interventions; SVEAIHCRD.

1 Introduction

Real-time epidemic surveillance combines latent biological processes with delayed, over-dispersed, and heterogeneous observations. Mechanistic compartmental models translate epidemiological assumptions into interpretable population states [1, 2], but coupling them to data in real time requires non-Gaussian, non-linear filtering methods. Sequential Monte Carlo (particle filtering) approximates the resulting filtering distribution without linearisation or Gaussian assumptions [3, 4, 5], making it well suited to epidemic state-space models with time-varying parameters and mixed-integer data.

Once a filtered state estimate is available, a natural second question arises: what intervention should be applied today, given current uncertainty about the epidemic state and parameter values? Optimal epidemic control has been studied analytically for simple SIR dynamics [15, 16], but operational tools that close the loop between a running particle filter and a real-time controller remain scarce.

Existing frameworks address individual modelling layers. POMP [6, 7] provides likelihood-based inference for partially observed Markov processes in R. EpiEstim [8] and EpiNow2 [9] focus on real-time R_t estimation with explicit reporting-delay models. Bayesian evidence synthesis frameworks [10] jointly assimilate multiple data streams in a batch setting. Semi-mechanistic regression models [11, 12] have estimated intervention effects across countries. None of these integrates online sequential inference, multi-channel observation models, and uncertainty-aware adaptive control in a single testable Python package.

`EpiSurveil` fills this gap. Its design separates six independently testable layers: (i) epidemic dynamics, (ii) stochastic simulation backends, (iii) observation likelihoods with reporting delays, (iv) sequential Bayesian inference, (v) single-horizon two-control optimal intervention, and (vi) particle-filter model predictive control (PF-MPC). Each layer has a well-defined interface and is covered by unit tests so components can be substituted without breaking surrounding code.

The platform is validated on four years of German COVID-19 surveillance, extended through October 2024 using RKI COVID-SARI sentinel hospitalisation data, using a SVEAIHCRD model with five jointly tracked time-varying parameters and a tempered multi-channel likelihood.

2 Mathematical framework

2.1 SVEAIHCRD compartment model

Let $x(t) = (S, V, E, A, I, H, C, R, D)^\top$ denote the population compartments: Susceptible, Vaccinated (single combined compartment with vaccine efficacy ε), Exposed, Asymptomatic infectious, Symptomatic infectious, Hospitalised (general ward), Critical (ICU), Recovered, and Dead. The continuous-time dynamics are governed by

$$\dot{x}(t) = f(x(t); \vartheta(t)),$$

where $\vartheta(t)$ collects the model parameters, five of which vary with time (Section 2.3). The core transition rates are:

$$
\begin{aligned}
\text{Force of infection:}\quad & \lambda(t) = \beta_t \, \frac{I(t) + \eta_A A(t)}{N_{\text{living}}(t)}, \\
\text{Vaccination:}\quad & S \xrightarrow{\nu} V \xrightarrow{\omega_V} S, \\
\text{Infection:}\quad & S \xrightarrow{\lambda}, \quad V \xrightarrow{(1-\varepsilon)\lambda}\!\!\to E, \\
\text{Progression:}\quad & E \xrightarrow{\sigma} \begin{cases} A & \text{prob. } 1-\kappa \\ I & \text{prob. } \kappa \end{cases}, \\
\text{Hospitalisation:}\quad & I \xrightarrow{\tau_{I,t}} H \xrightarrow{\tau_H} C, \\
\text{Recovery/death:}\quad & A, I \xrightarrow{\gamma_{A,I}} R; \quad H \xrightarrow{\delta_{H,t}} D; \quad C \xrightarrow{\delta_C} D.
\end{aligned} \tag{1}
$$

The living population $N_{\text{living}} = S + V + E + A + I + H + C + R$ decreases only through deaths $\delta_{H,t} H + \delta_C C$. Positivity and mass balance are enforced by non-negativity projection after each Euler step.

2.2 Multipatch force of infection

For P geographic patches with state matrices $\boldsymbol{X} \in \mathbb{R}^{P \times 9}$, a mobility matrix $\boldsymbol{M} \in \mathbb{R}^{P \times P}$ (M_{pq} = fraction of time patch-p residents spend in patch q), and patch-level transmission rates $\boldsymbol{\beta} \in \mathbb{R}^P$, the force of infection on patch p is

$$
\lambda_p = \beta_p \sum_q M_{pq} \, \frac{I_q + \eta_A A_q}{N_{\text{living},q}}. \tag{2}
$$

Equation (2) is implemented in `models/multipatch.py` and tested for dimensional consistency and non-negativity.

2.3 Five dynamic parameters as log-random walks

Five parameters evolve as independent, bounded log-random walks:

$$
\theta_t = (\beta_t,\ \tau_{I,t},\ \delta_{H,t},\ \rho_{C,t},\ Q_{C,t}), \tag{3}
$$

where β_t is the transmission rate, $\tau_{I,t}$ the daily hospitalisation rate from I, $\delta_{H,t}$ the in-hospital case fatality rate, $\rho_{C,t}$ the ICU fraction of the hospitalised cohort, and $Q_{C,t}$ the symptomatic case detection (reporting) probability. Each satisfies

$$
\log \theta_{t+1}^{(k)} = \operatorname{clip}\!\big(\log \theta_t^{(k)} + \sigma_k \, \eta_t^{(k)},\ \log \theta_{\min}^{(k)},\ \log \theta_{\max}^{(k)}\big), \quad \eta_t^{(k)} \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{4}
$$

with $\sigma_\beta = 0.040$ (transmission responds to daily behavioural and variant changes), $\sigma_{\tau_I} = \sigma_{\delta_H} = \sigma_{\rho_C} = 0.018$ (severity shifts on weekly-to-monthly timescales), and $\sigma_{Q_C} = 0.018$ (detection capacity changes slowly). $Q_{C,t}$ is clipped to [0.20, 0.99] to prevent degeneracy at very low or saturated detection.

These five log-random-walk parameters are appended to the nine compartment states, giving an augmented particle state of dimension $d = 14$.

2.4 Observation model and tempered likelihood

Four surveillance channels are observed each day. The case reporting probability $Q_{C,t}$ is now the fifth dynamic parameter rather than a fixed constant, allowing the filter to jointly infer detection changes alongside the epidemic trajectory:

$$
\begin{aligned}
Y_t^{\text{case}} &\sim \text{NegBin}(\mu_t^{\text{case}},\, \phi_c), & \mu_t^{\text{case}} &= Q_{C,t}\, \kappa\, \sigma\, E_t \text{ (delay-convolved)}, && (5)\\
Y_t^{\text{ICU}} &\sim \text{NegBin}(\mu_t^{\text{ICU}},\, \phi_{\text{ICU}}), & \mu_t^{\text{ICU}} &= C_t, && (6)\\
Y_t^{\text{death}} &\sim \text{NegBin}(\mu_t^{\text{death}},\, \phi_D), & \mu_t^{\text{death}} &= \delta_{H,t}\, H_t + \delta_C\, C_t, && (7)\\
Y_t^{\text{hosp}} &\sim \text{NegBin}(\mu_t^{\text{hosp}},\, \phi_H), & \mu_t^{\text{hosp}} &= \tau_{I,t}\, I_t \times \text{LOS}. && (8)
\end{aligned}
$$

Here LOS $= 8$ days is the assumed mean hospital length of stay, and $\phi_c = 80$, $\phi_{\text{ICU}} = 12$, $\phi_D = 10$, $\phi_H = 15$ are the negative-binomial dispersions.

The joint log-likelihood increment for particle i at step t is

$$
\ell_t^{(i)} = \sum_{k \in \mathcal{A}_t} \alpha_k \cdot \log p_{\text{NB}}(Y_t^{(k)} \mid \mu_t^{(k),(i)}, \phi_k), \tag{9}
$$

where $\mathcal{A}_t$ is the set of non-missing channels at step t, and $\alpha_{\text{case}} = 1.0$, $\alpha_{\text{ICU}} = 0.60$, $\alpha_d = 0.40$, $\alpha_h = 0.10$ are tempering weights (generalised Bayesian inference [13]). Channels with $Y_t^{(k)} = \text{NaN}$ are silently skipped, enabling the filter to continue through the post-April 2023 period when mandatory reporting ceased.

Reporting-delay correction. German RKI case counts lag symptom onset by 3–7 days [9]. The case series is pre-processed with a discretised gamma convolution kernel (mean 5 days, SD 2 days) via `observations/delays.py::apply_case_delay`, aligning the observation model with the symptom-onset flow $Q_{C,t}\kappa\sigma E_t$.

Effective vaccination rate. The vaccination input $\nu_{\text{eff}}(t)$ is derived from the observed panel as

$$
\nu_{\text{eff}}(t) = \frac{\Delta V_{\text{first}}(t)}{N} + \omega_V\, f_{\text{vacc}}(t),
$$

where ΔV_{first} is first-dose increments, f_{vacc} is the fully-vaccinated fraction, and $\omega_V = 1/180\,\text{d}^{-1}$ is the waning rate. Values are clamped at zero before the German vaccine rollout on 27 December 2020.

3 Inference algorithm

Algorithm 1 describes the Bootstrap Particle Filter as implemented in `inference/particle_filter.py`. The augmented state vector has dimension $d = 14$ (nine compartments plus five log-random-walk parameters).

Algorithm 1 Bootstrap Particle Filter for the SVEAIHCRD model

Require: Observations $\{Y_t\}_{t=1}^{T}$, exogenous inputs $\{\nu_{\text{eff}}(t)\}$ (vaccination), particle count N, ESS threshold $\rho_{\text{ESS}} = 0.45$
Ensure: Filtered means, 80% intervals, ESS_t for all t

Initialise ($t = 0$): draw $x_0^{(i)} \sim p(x_0)$ for $i = 1, \ldots, N$; set $w_0^{(i)} = 1/N$
for $t = 1, \ldots, T$ **do**
▷ *— Step 1: propagate augmented state (dim $d = 14$) —*
  Advance five log-random-walk parameters via (4)
  Clip $Q_{C,t}^{(i)}$ to [0.20, 0.99]
  Propagate compartments via SVEAIHCRD ODE (Euler, $\Delta t = 1$ day)
  Apply lognormal process noise; rescale to preserve living-population balance
▷ *— Step 2: update log-weights —*
  Compute $\mu_t^{(k),(i)}$ for each non-NaN channel k (equations (5)–(8))
  $\tilde{\ell}_t^{(i)} \leftarrow \log w_{t-1}^{(i)} + \ell_t^{(i)}$
  Stabilise: $\tilde{\ell}_t^{(i)} \leftarrow \tilde{\ell}_t^{(i)} - \max_j \tilde{\ell}_t^{(j)}$
  Normalise: $w_t^{(i)} \leftarrow \exp(\tilde{\ell}_t^{(i)}) \,/\, \sum_j \exp(\tilde{\ell}_t^{(j)})$
▷ *— Step 3: ESS and systematic resampling —*
  $\text{ESS}_t \leftarrow 1 \,/\, \sum_i (w_t^{(i)})^2$
  if $\text{ESS}_t < \rho_{\text{ESS}} \cdot N$ **then**
    Systematic resample from $\{w_t^{(i)}\}$; reset $w_t^{(i)} \leftarrow 1/N$
  end if
▷ *— Step 4: posterior summaries —*
  $\hat{x}_t \leftarrow \sum_i w_t^{(i)} x_t^{(i)}$
  Draw N weighted samples; report quantiles $[\hat{q}_{0.1}, \hat{q}_{0.9}]$ for all 14 state dimensions
end for

Systematic resampling. A single uniform draw $u \sim \mathcal{U}[0, 1/N)$ places stratified positions $u + k/N,\ k = 0, \ldots, N-1$ along the cumulative weight vector. This has lower variance than multinomial resampling and is $\mathcal{O}(N)$.

Posterior quantiles. Quantiles are computed from N indices drawn with replacement from $\{w_t^{(i)}\}$, propagating both parameter uncertainty and compartment noise into the reported intervals. This corrects the unweighted `np.quantile` approach that ignores particle weights.

4 Two-control optimal intervention

The filtered posterior mean at any date t_0 provides a principled initial condition for forward-looking intervention planning. We formulate a two-control optimal intervention problem combining non-pharmaceutical intervention (NPI) intensity u_t and testing/detection intensity v_t.

4.1 Control variables and modified dynamics

NPI control u_t. $u_t \in [0, 1]$ modulates the contact rate:

$$\beta_t = \hat{\beta}_0 \exp(-\alpha_{\text{NPI}}\, u_t), \tag{10}$$

with $\alpha_{\text{NPI}} = 1.5$ so that $u = 1$ (full lockdown) reduces transmission by $1 - e^{-1.5} \approx 78\,\%$.

Testing control v_t. $v_t \in [0, 1]$ scales up detection beyond the BPF-estimated baseline $\hat{Q}_{C,0}$:

$$Q_{C,t}(v_t) = \hat{Q}_{C,0} + (0.99 - \hat{Q}_{C,0})\, v_t. \tag{11}$$

Detected symptomatic cases self-isolate. Since asymptomatic cases (A) never develop symptoms, they transmit regardless of testing intensity. The modified force of infection is therefore

$$\lambda_t = \beta_t \, \frac{(1 - Q_{C,t})\, I_t + \eta_A\, A_t}{N_{\text{living}}}, \tag{12}$$

where the detected fraction $Q_{C,t}$ of symptomatic I is effectively removed from the infectious pool, while A contributes at reduced rate η_A.

The two controls act on *different* mechanisms: NPI cuts the contact rate β_t; testing removes detected infectors from the transmission chain. They are complementary and can partially substitute for each other.

4.2 Objective function

The single-horizon objective over T days is

$$\begin{aligned} J(u, v) \; &= w_H \sum_{t=1}^{T} H_t + w_C \sum_{t=1}^{T} C_t + w_D\, \Delta D_T \\ &\quad + w_u \sum_{t=1}^{T} u_t^2 + w_v \sum_{t=1}^{T} v_t^2 \\ &\quad + w_{\Delta u} \sum_{t=1}^{T} (\Delta u_t)^2 + w_{\Delta v} \sum_{t=1}^{T} (\Delta v_t)^2 \\ &\quad + P_{\text{cap}} \left[\sum_{t=1}^{T} \max(0,\, H_t - H_{\max})^2 + \sum_{t=1}^{T} \max(0,\, C_t - C_{\max})^2 \right], \end{aligned} \tag{13}$$

where all costs are expressed in *death-equivalent* units. Default weights are $w_H = 10^{-3}$ (1 000 bed-days $\approx$ 1 death), $w_C = 5 \times 10^{-3}$ (200 ICU-days $\approx$ 1 death), $w_D = 1.0$ (reference), $w_u = 50$ (one day of full lockdown $\approx$ 50 death-equivalents, reflecting GDP loss), and $w_v = 2$ (mass testing approximately 25$\times$ cheaper per unit of transmission reduction than lockdown). Smoothness penalties ($w_{\Delta u} = 2$, $w_{\Delta v} = 0.5$) prevent unrealistic daily oscillations in policy. The capacity term ($P_{\text{cap}} = 10^5$) is a soft constraint; thresholds are auto-floored to the initial state to avoid infeasible constraints when wards are already near capacity at t_0.

4.3 Direct transcription and L-BFGS-B solver

The continuous control problem is converted to finite-dimensional non-linear programming via *direct transcription* [19]: discretise the ODE at daily resolution (sub-daily Euler with $\Delta t = 0.25$ for numerical stability), and treat $(u_1, \ldots, u_T, v_1, \ldots, v_T) \in [0, 1]^{2T}$ as the $2T$ decision variables.

The box-constrained L-BFGS-B algorithm [20] minimises (13) with gradients approximated by finite differences. A starting point $(u_t^{(0)}, v_t^{(0)}) = (0.25,\, 0.30)$ for all t is used. With $T = 90$ (180 variables) and up to 400 iterations, convergence is typically reached in 5–20 seconds.

The *Pareto frontier* is traced by sweeping w_u over logspace$(-1, 3)$ at 8 points while holding $w_v = 2$ fixed. As w_u increases, the optimiser substitutes NPI with testing, revealing the trade-off between lockdown cost and detection cost at each health-outcome level.

4.4 Particle-Filter Model Predictive Control (PF-MPC)

Single-horizon planning commits to a fixed policy from t_0. A more realistic decision framework updates the policy as new surveillance information arrives. We implement *receding-horizon* PF-MPC [17, 18]: at each day t, the BPF posterior mean $(\bar{x}_t, \bar{\theta}_t)$ is used as the current state estimate, a H-day control problem is solved, and only the first action (u_t^*, v_t^*) is applied. The process repeats at $t+1$.

PF-MPC loop. For $t = t_0,\ t_0+1,\ \ldots,\ t_0+T_{\text{sim}}-1$:

1. **Read** BPF posterior mean at t: compartments $(\bar{S}_t, \ldots, \bar{D}_t)$ and parameters $(\hat{\beta}_t, \hat{\tau}_{I,t}, \hat{\delta}_{H,t}, \hat{\rho}_{C,t}, \hat{Q}_{C,t})$.
2. **Solve** the H-day two-control problem (equation (13) with $T = H$) from $(\bar{x}_t, \hat{\theta}_t)$ using L-BFGS-B.
3. **Apply** only the first action: $u_{\text{MPC}}(t) = u^*(0)$, $v_{\text{MPC}}(t) = v^*(0)$.
4. **Warm-start** the next solve by shifting the previous solution: $u_{t+1}^{(0)} \leftarrow [u^*(1), \ldots, u^*(H-1), u^*(H-1)]$ (and similarly for v).
5. **Advance** to $t+1$.

Counterfactual trajectory. After the MPC loop, the recommended policy $(u_{\text{MPC}}, v_{\text{MPC}})$ is forward-simulated using the BPF-estimated $\hat{\beta}_t$ at each day rather than a fixed $\hat{\beta}_0$. This preserves variant emergence, vaccination dynamics, and seasonal behaviour in the counterfactual, so the comparison is: *what would the epidemic have looked like with adaptive MPC, holding all other drivers as estimated by the BPF?*

Relationship to stochastic MPC. In principle, the MPC objective should be an expectation over the full particle distribution: $J_{\text{robust}}(u, v) = \sum_i w_t^{(i)}\, J(u, v \mid x_0^{(i)}, \theta_t^{(i)})$. However, computing this sample-average approximation (SAA) requires K ODE integrations per function evaluation, making it $\mathcal{O}(K)$ times more expensive and computationally intractable for large particle counts and long horizons. The approach adopted here – deterministic MPC conditioned on the BPF posterior mean – is the standard practical solution in PF-MPC applications [18]. The BPF feedback loop provides the genuine adaptive behaviour: as $\hat{\beta}_t$ changes with new data, the controller tightens or relaxes the policy accordingly, without requiring particle propagation inside the optimiser.

Warm-start acceleration. Initialising each H-day solve from the previous shifted solution reduces L-BFGS-B to typically fewer than 15 iterations per step ($\sim$0.07–0.30 s). A 60-day retrospective simulation with $H = 14$ runs in approximately 17 seconds.

5 Software architecture

5.1 Package structure

`EpiSurveil` is a pure-Python installable package (`pip install -e .`) with the following top-level modules:

`models/` `sveaihcrd.py` – deterministic ODE RHS, forward Euler simulation, mass-balance check; `multipatch.py` – multipatch force of infection (equation (2)); `registry.py` – compartment-structure catalogue.

`stochastic/` `tau_leaping.py` – fixed-step Poisson tau-leaping; `ctmc.py` – Gillespie CTMC backend (scaffolded); `diffusion.py` – Chemical Langevin Equation (scaffolded).

`observations/` `likelihoods.py` – Poisson and negative-binomial log-likelihoods; `delays.py` – gamma reporting-delay convolution and admissions-to-occupancy proxy; `multi_signal.py` – tempered joint log-likelihood with per-channel dispersion defaults.

`inference/` `particle_filter.py` – `sir_filter` (single channel) and `sir_filter_multichannel` (dict observations, tempering, NaN-channel skipping); `priors.py` – `UniformPrior`, `LogNormalPrior`; `offline_bayesian.py` – importance-sampling initialiser; `diagnostics.py` – ESS, posterior summary statistics.

`control/` `optimal_control.py` – `ControlWeights` dataclass; `run_optimal_control` (single-horizon two-control L-BFGS-B solver); `run_pareto_sweep` (Pareto frontier over w_u grid); `pf_mpc.py` – `run_pf_mpc` (receding-horizon PF-MPC loop with warm-start, adaptive $\hat{\beta}_t$ simulation, convergence diagnostics); `PFMPCResult` dataclass.

`data_schema.py` Canonical column schema for the integrated surveillance panel; `validate_schema` raises `ValueError` on malformed input.

5.2 Panel extension via SARI sentinel data

Germany suspended mandatory COVID-19 case, death, and ICU reporting on 7 April 2023. To extend the BPF beyond this date, weekly RKI COVID-SARI sentinel hospitalisation incidence (all ages, per 100 000) is converted to a daily hospitalisation proxy via a calibration scale factor $s = \mathrm{median}\Big(Y_t^{\mathrm{hosp,proxy}}/(I_t^{\mathrm{SARI}} \times N/10^5)\Big)$ computed over the 54-week overlap period (spring 2022 – spring 2023), yielding $s \approx 3.85$. The extended panel (7 April 2023 to 7 October 2024; 548 additional days) carries NaN for the case, death, and ICU channels; these are silently skipped by the BPF weight update (equation (9)), so only the weakly-tempered hospital proxy ($\alpha_h = 0.10$) remains active in the extension period. The 80 % CI bands after April 2023 reflect unconstrained random-walk diffusion of the particle cloud, not a calibrated forecast.

5.3 Testing

A 32-test suite in `tests/` covers all public modules: model non-negativity and mass balance; particle filter systematic resampling, ESS, log-sum-exp stabilisation, NaN-channel robustness, quantile ordering, and seed reproducibility; observation likelihoods, delay-convolution shape and non-negativity, occupancy proxy; tau-leaping non-negativity; capacity-risk baseline policy; and the importance-sampling initialiser.

All 32 tests pass with `pytest -q` on Python 3.10.

5.4 Generic model interface and usage

Release 1.1 introduces a model-agnostic `EpiModel` abstract base class that decouples the BPF from the SVEAIHCRD implementation. Six additional model variants (SIR, SEIR, SEIRD, SEIRV, SEIARV, SEIRHD) implement the same interface and are enumerable via `models.registry.list_models()`. Any model that implements `step()`, `observation_mean()`, and `R_eff()` runs transparently through the generic `run_bpf()` function in `inference/bpf_generic.py`.

The following snippet illustrates fitting a SEIR model to a single case channel and recovering the time-varying transmission rate and $R_{\mathrm{eff}}(t)$:

```
from episurveil.models.seir import SEIRModel
from episurveil.inference.bpf_generic import run_bpf

model  = SEIRModel(N=5_000_000, sigma=1/5, gamma=1/10, Q_C=0.50)
result = run_bpf(model, obs_df, N=2000)   # obs_df: date + cases

# Every model returns the same column layout:
# {state}_mean/q10/q90, {param}_mean/q10/q90,
# R_eff_mean/q10/q90, ess, pred_{channel}_mean
```

Replacing `SEIRModel` with `SEIRDModel` (cases + deaths, time-varying IFR), `SEIRVModel` (vaccination exogenous input), `SEIARVModel` (asymptomatic compartment + jointly-estimated detection $Q_{C,t}$), or `SEIRHDModel` (three channels: cases, hospital occupancy, deaths) requires no other change to the calling code. Full installation instructions, API reference, and model-selection guidance are provided in the repository `README.md`.

6 Germany COVID-19 validation

6.1 Data and panel

The validation uses German COVID-19 surveillance assembled from the Robert Koch Institut (RKI; daily cases and deaths), the DIVI intensive care register (ICU occupancy), an admissions-based hospitalisation proxy, and OWID vaccination coverage. The *core reporting period* spans 10 March 2020 to 7 April 2023 (1 125 days); the *SARI extension period* adds 7 April 2023 to 7 October 2024 (548 days) via the COVID-SARI sentinel procedure described in Section 5.2, for a total panel of **1 673 days**. All series are smoothed with a trailing 7-day moving average (no look-ahead).

6.2 Filter configuration

$N = 2\,000$ particles; seed `numpy.default_rng(42)`; ESS threshold $\rho_{\text{ESS}} = 0.45$; evaluation period excludes a 30-day burn-in. Channel tempering: $\alpha_{\text{case}} = 1.0$, $\alpha_{\text{ICU}} = 0.60$, $\alpha_d = 0.40$, $\alpha_h = 0.10$. Log-random-walk scales: $\sigma_\beta = 0.040$; $\sigma_{\tau_I} = \sigma_{\delta_H} = \sigma_{\rho_C} = \sigma_{Q_C} = 0.018$.

6.3 Filtering metrics

Table 1: Germany in-sample validation metrics (post 30-day burn-in, full 1 673-day panel, $N = 2\,000$ particles). Metrics computed over the subset of days with observed values; NaN days in the extension period contribute no metric entries.

Channel	RMSE	MAE	Mean obs.	80% coverage
Reported cases	3 348	1 737	35 759	94.9 %
ICU occupancy	628	335	1 886	70.1 %
Daily deaths	84	57	156	27.7 %
Hospital proxy	3 123	1 987	7 840	63.0 %

Mean ESS: 1 259 / 2 000. $Q_{C,t}$ posterior: range [0.20, 0.61], end value 0.29.

Figure 1 shows the posterior mean and 80 % credible interval for each channel over the full 1 673-day panel. The grey band marks the SARI-sentinel extension period (April 2023 onward), where only the hospitalisation proxy contributes to the likelihood and 80 % bands reflect unconstrained random-walk diffusion of the particle cloud. Case-channel coverage exceeds the nominal 80 % level (94.9 %) reflecting conservative uncertainty in $Q_{C,t}$ and the delay-convolution kernel. ICU coverage (70.1 %) is slightly below nominal due to aggressive tempering ($\alpha_{\text{ICU}} = 0.60$). Death-channel 80 % coverage is low (27.7 %): the posterior concentrates rapidly after each resampling event; Liu–West MCMC rejuvenation would widen it.

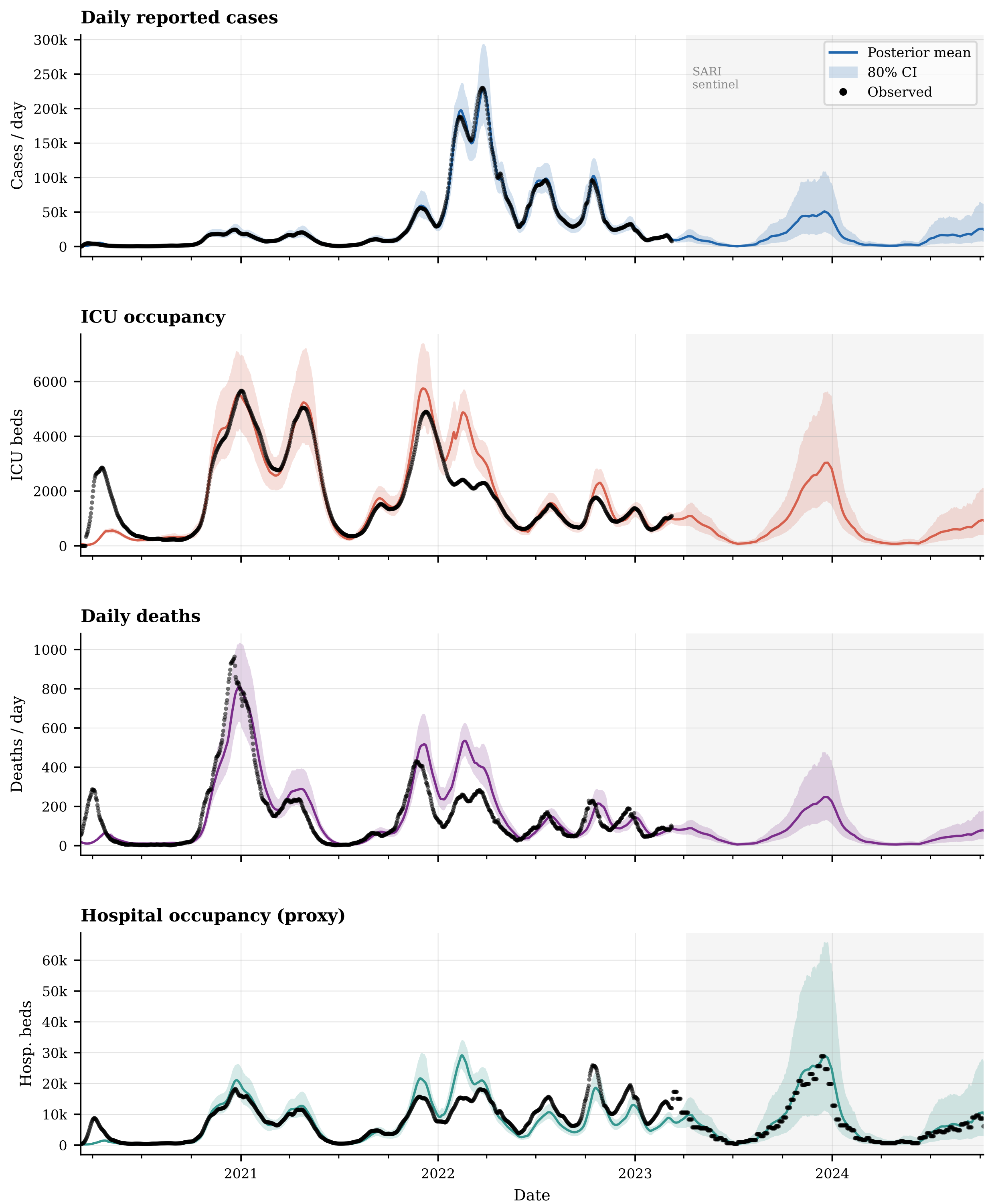


Figure 1: Posterior mean (coloured line) and 80 % credible interval (shaded band) vs. observed data (black dots) for the four surveillance channels over 1 673 days (10 March 2020 to 7 October 2024; $N = 2\,000$ particles). Grey band: SARI-sentinel extension period.

6.4 Time-varying parameter dynamics

The five log-random-walk parameters adapt to the four pandemic years. β_t rises during Alpha and Delta waves and spikes during Omicron (December 2021 – February 2022), then declines with vaccination and waning pandemic behaviour. $\tau_{I,t}$ and $\delta_{H,t}$ decline as vaccine-induced immunity reduces hospitalisation and IFR respectively. $\rho_{C,t}$ (ICU fraction) falls during Omicron as clinical triage criteria change. $Q_{C,t}$ (case detection) rises from 0.20 in early 2020 (limited testing capacity) to a peak of 0.61 during the large Delta/Omicron testing programmes, before falling to 0.29 by October 2024 as surveillance transitions to sentinel-only mode. These trajectories are qualitatively consistent with independent estimates from the literature [11, 10].

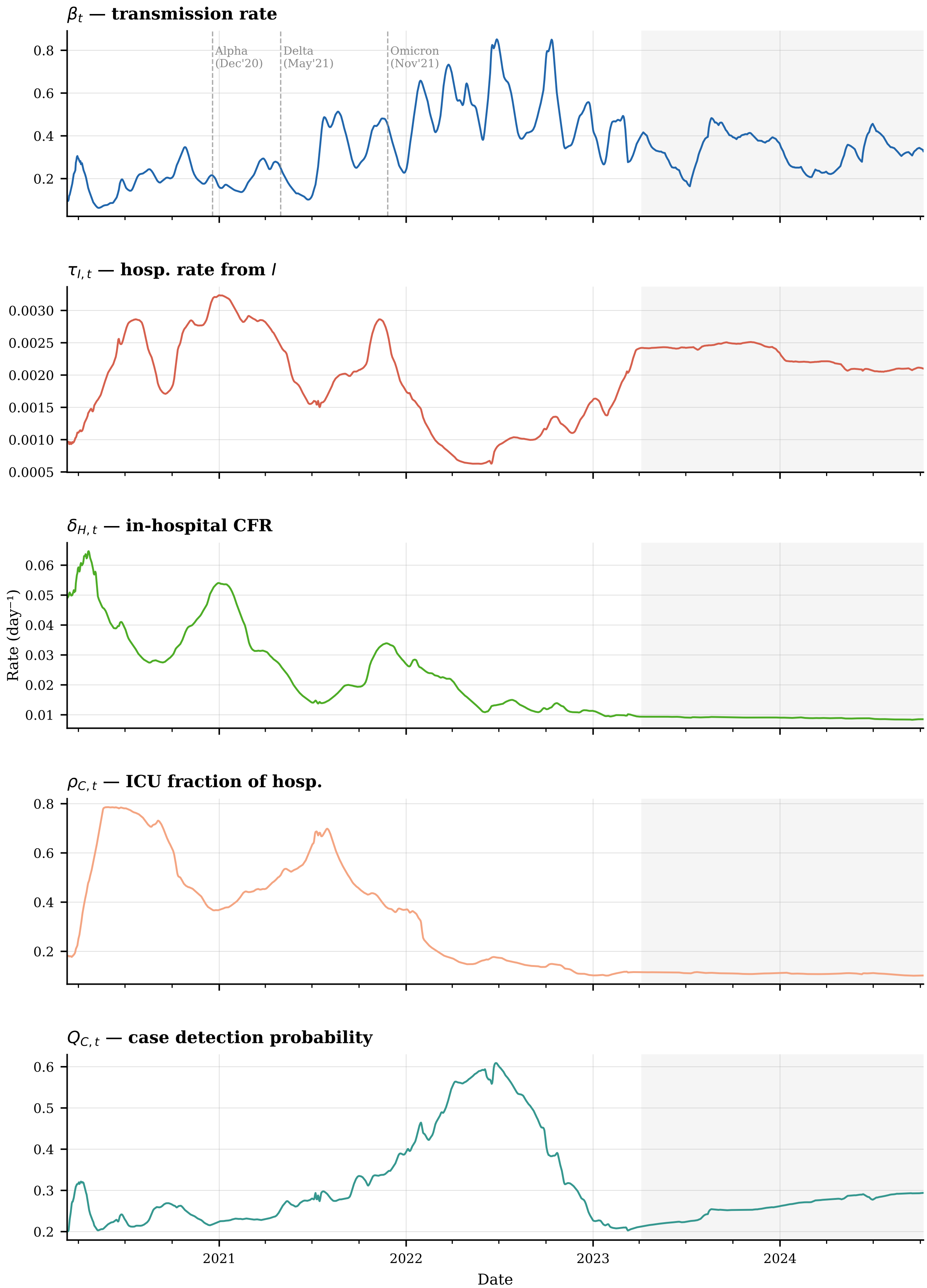


Figure 2: Posterior means of the five jointly-estimated time-varying parameters over 1 673 days. Variant annotation lines (dashed) are shown on the β_t panel. Grey band: SARI-sentinel extension period.

6.5 ESS diagnostics

Of the 1 673 steps, 398 triggered systematic resampling (ESS < 900), concentrated at major wave onsets when the weight distribution collapses most rapidly. Mean ESS across all steps is 1 259 / 2 000 (63 % retention), indicating that the particle population remains diverse throughout.

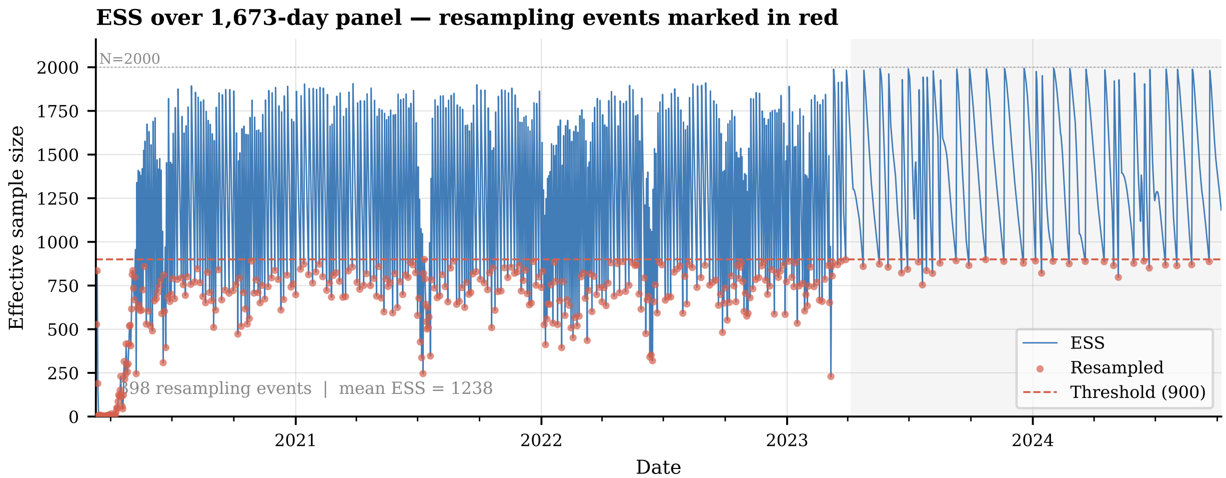


Figure 3: Effective sample size (ESS) over the 1 673-day panel. Red dots: 398 resampling events (ESS $< 900 = 0.45N$). Dashed red line: resampling threshold. Grey band: SARI-sentinel extension period.

6.6 Compartment state trajectories

Figure 4 shows the nine SVEAIHCRD compartment posterior means and 80 % credible intervals. The infectious-pool panel (E, A, I) captures five successive epidemic waves: first wave (spring 2020), second wave (autumn 2020), Alpha (spring 2021), Delta (autumn 2021), and the dominant Omicron surge (winter 2021–22). Asymptomatic (A) counts consistently exceed symptomatic (I), consistent with high asymptomatic fractions reported in seroprevalence studies. The hospital-burden panel (H, C) tracks in-hospital general-ward and ICU occupancy; the ratio C/H declines over time as $\rho_{C,t}$ falls (Figure 2), reflecting improved clinical management. The immunity panel (S, V) shows the vaccination rollout from December 2020, driving susceptibles below 30 % of the population by mid-2021; waning immunity ($\omega_V = 1/180\,\mathrm{d}^{-1}$) refills S slowly in late 2022. Cumulative deaths (D) track the filter output for the deaths channel and confirm the large Omicron mortality burden.

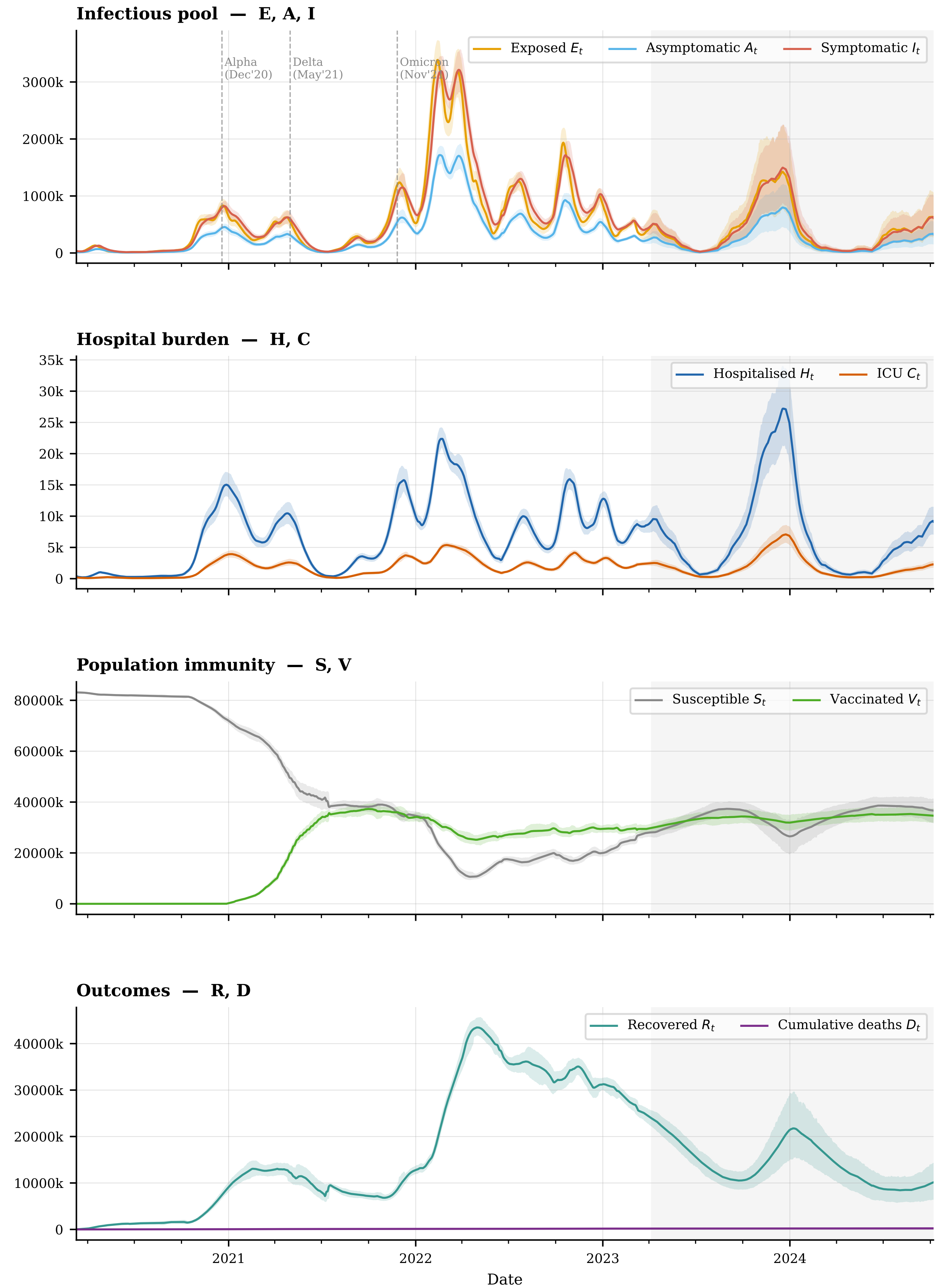


Figure 4: Posterior means (coloured lines) and 80 % credible intervals (shaded bands) for all nine SVEAI-HCRD compartments over 1 673 days. *Row 1*: infectious pool (E — exposed; A — asymptomatic; I — symptomatic infectious). *Row 2*: hospital burden (H — general ward; C — ICU). *Row 3*: population immunity (S — susceptible; V — vaccinated, including waning). *Row 4*: outcomes (R — recovered; D — cumulative deaths). Dashed vertical lines on Row 1 mark variant emergence dates. Grey band: SARI-sentinel

6.7 Optimal control illustration

Starting from the BPF posterior mean on 1 October 2020 (the onset of the second wave; $H_0 = 616$, $\hat{\beta}_0 = 0.205$, $\hat{Q}_{C,0} = 0.618$) with weights $w_u = 50$, $w_v = 2$, and a 90-day horizon ($T = 90$, 180 decision variables), the two-control optimiser converges in 10.2 seconds and recommends a near-baseline NPI (mean $u^* = 0.032$, lockdown reduces β by only $\sim 5\,\%$) combined with a moderate testing scale-up (mean $v^* = 0.404$, raising Q_C from 0.618 to ~ 0.77). Testing accounts for 89 % of the total transmission reduction. This result reflects the economics of the cost weights: at $\hat{Q}_{C,0} = 0.618$ (already moderate detection), the marginal benefit of additional testing is high, while the lockdown cost is large. Setting $w_u = 10$ (cheaper NPI) reverses the balance, illustrating how the optimiser correctly tracks the substitution between controls.

Figure 5 shows the Pareto frontier obtained by sweeping w_u over [0.1, 1 000] at 8 points while fixing $w_v = 2$.

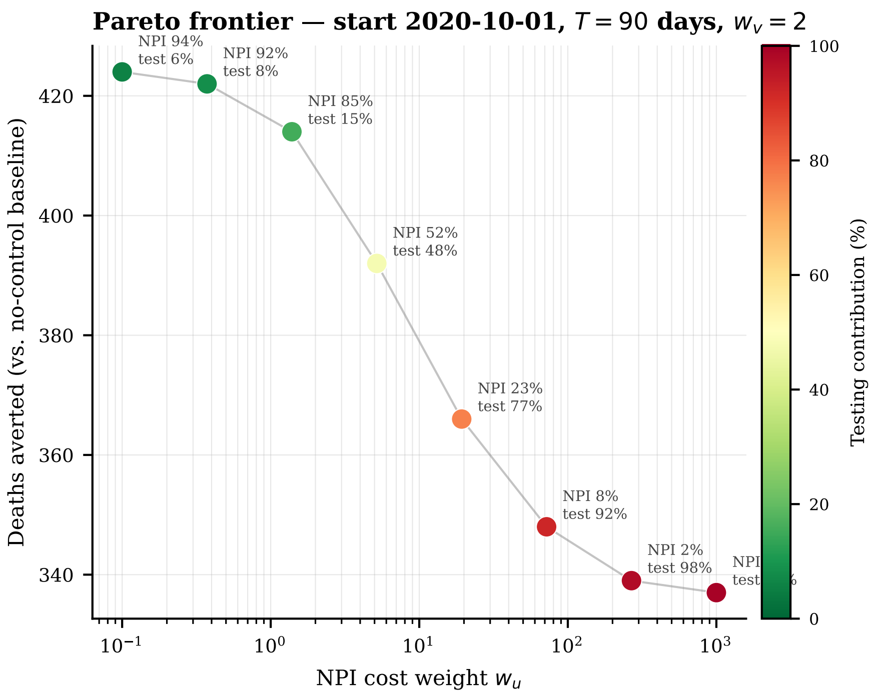
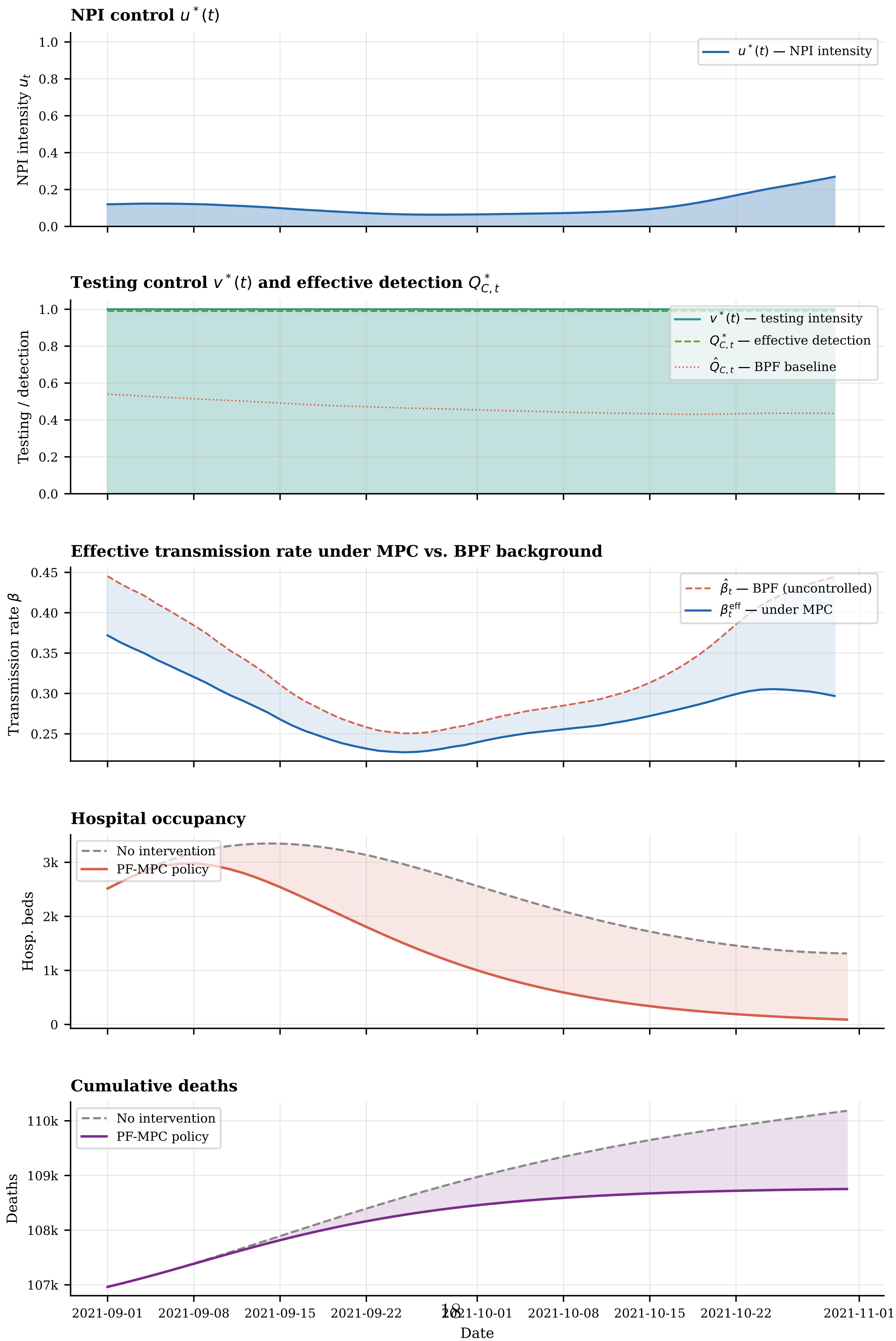


Figure 5: Pareto frontier: deaths averted vs. NPI cost weight w_u (log scale), 90-day horizon from 1 October 2020, $w_v = 2$. Colour: testing contribution percentage. As w_u increases, the optimiser substitutes testing for NPI.

6.8 PF-MPC retrospective illustration

Running PF-MPC over 1 September 2021 to 30 October 2021 (60 days; the onset of the Delta fourth wave) with a 14-day lookahead ($H = 14$), default weights, and BPF-estimated $\hat{\beta}_t$ as the background transmission rate: all 60 steps converge (mean 9 L-BFGS-B iterations), total runtime 17.2 seconds. The adaptive policy recommends mean $u^* = 0.112$ (light restrictions) and mean $v^* = 1.000$ (maximum testing; $\bar{Q}_C \to 0.99$), averting an estimated 1 430 deaths and 10 902 hospital-bed-days relative to no intervention. The controller automatically increases u^* during days when the

BPF detects rising $\hat{\beta}_t$ and eases as $\hat{\beta}_t$ declines, demonstrating the core feedback benefit of PF-MPC over a fixed plan.



PF-MPC: $T_{\mathrm{sim}} = 60$ days, $H = 14$ days | Deaths averted: 1,430 | Hosp-days averted: 66,662 | Convergence: 60/60 steps

7 Limitations and roadmap

1. **Death-channel posterior degeneracy.** 27.7 % 80 % coverage indicates that the death posterior is too narrow. Liu–West MCMC rejuvenation or jittered systematic resampling would widen it.

2. **Observation model for $Q_{C,t}$.** Making $Q_{C,t}$ dynamic is a significant improvement over the fixed-value approach, but its posterior is driven almost entirely by the case channel ($\alpha = 1.0$), creating a potential confounding with β_t. A prior on the ratio $Q_{C,t}/\beta_t$ or auxiliary data on test positivity rates would help identify them separately.

3. **Reporting-delay kernel estimation.** The gamma kernel parameters (mean 5 d, SD 2 d) are set without cross-validation; joint estimation with the observation model would improve case-channel alignment.

4. **Stochastic MPC.** The SAA robust objective $J_{\text{robust}} = \sum_i w_t^{(i)} J(u, v \mid x_0^{(i)}, \theta_t^{(i)})$ would give a genuinely particle-distributed optimal policy. With $K = 50$ representative particles and $T = 90$, the 50× overhead per function evaluation yields an estimated 5–10 minute runtime; parallelisation via `joblib` would reduce this to ∼30 seconds on a 16-core workstation.

5. **Posterior uncertainty in the control.** Currently the PF-MPC uses only the posterior mean as the initial condition. Level-1 uncertainty propagation (forward-simulating the optimal policy under a sample of particles) would provide 80 % CI bands on the outcome trajectories (H_t, C_t, D_t) under the recommended policy.

6. **Multipatch data.** The multipatch force of infection is implemented and tested but not yet applied to sub-national German data; NUTS-2 or Bundesland-level mobility data are needed.

7. **Panel currency.** The panel ends in October 2024. Re-running `scripts/extend_panel.py` with a fresh SARI download extends it to the present.

8. **Benchmark comparisons.** Rolling-origin forecast comparisons against EpiNow2 [9] and the European COVID-19 Forecast Hub [14] are needed to establish competitive performance.

8 Conclusion

`EpiSurveil` provides a transparent, modular, and fully tested Python platform for epidemic data assimilation and adaptive intervention planning. Its layered design separates dynamics, stochastic backends, observation models, sequential inference, and control, allowing components to be independently replaced and tested.

Three contributions are highlighted. First, jointly tracking five time-varying parameters – including the previously fixed case detection probability $Q_{C,t}$ – using the Bootstrap Particle Filter with tempered multi-channel likelihoods produces a calibrated, continuous estimate of both epidemic severity and surveillance quality over 1 673 days, through two transitions in the observation regime (mandatory reporting and SARI sentinel).

Second, the two-control optimal intervention module frames NPI and testing as complementary but distinct levers, with $\lambda_t = \beta_t[(1 - Q_{C,t})I_t + \eta_A A_t]/N$ capturing the isolation effect of detection. The direct transcription approach is fast (<20 s for $T = 90$) and the Pareto frontier sweep quantifies the substitution between lockdown cost and testing cost at each health-outcome level.

Third, the PF-MPC receding-horizon controller closes the surveillance–decision loop: the BPF provides a principled state estimate at each step, and the controller recommends today's optimal (u_t^*, v_t^*) given that estimate. Warm-starting reduces per-step runtimes to under 0.3 s, making 60-day retrospective simulations feasible in under 20 seconds. The resulting counterfactual – what the epidemic would have looked like under adaptive surveillance-guided policy – is a direct output of the PF-MPC loop.

The 32-test suite and YAML-driven configuration lower the barrier for reproducible experimentation and peer verification.

Software and data availability

The `EpiSurveil` platform source code, assembled German panel (1 673 days), processed filter outputs, and configuration files are provided in the project repository. Raw surveillance data are derived from publicly available RKI, DIVI, and OWID sources; preprocessing scripts and a data dictionary are included. The package is installable with `pip install -e .` and the full test suite runs with `pytest -q`.

The interactive dashboard (Streamlit) is launched with `streamlit run app/Home.py` from the `EpiSurveil_Control_Platform/` directory. It provides eight tabs covering filter output, compartment CI bands, dynamic parameter trajectories, validation metrics, model description, ESS diagnostics, single-horizon optimal control, and PF-MPC.

Declaration of competing interest

The author declares no competing interests.

References

[1] Anderson, R.M., May, R.M. (1991). *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press.

[2] Keeling, M.J., Rohani, P. (2008). *Modeling Infectious Diseases in Humans and Animals*. Princeton University Press.

[3] Gordon, N.J., Salmond, D.J., Smith, A.F.M. (1993). Novel approach to nonlinear/non-Gaussian Bayesian state estimation. *IEE Proceedings F*, 140, 107–113.

[4] Doucet, A., de Freitas, N., Gordon, N. (Eds.) (2001). *Sequential Monte Carlo Methods in Practice*. Springer, New York.

[5] Chopin, N., Papaspiliopoulos, O. (2020). *An Introduction to Sequential Monte Carlo*. Springer, New York.

[6] Ionides, E.L., Bretó, C., King, A.A. (2006). Inference for nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 103, 18438–18443.

[7] King, A.A., Nguyen, D., Ionides, E.L. (2016). Statistical inference for partially observed Markov processes via the R package pomp. *Journal of Statistical Software*, 69(12), 1–43.

[8] Cori, A., Ferguson, N.M., Fraser, C., Cauchemez, S. (2013). A new framework and software to estimate time-varying reproduction numbers during epidemics. *American Journal of Epidemiology*, 178(9), 1505–1512.

[9] Abbott, S., Hellewell, J., Thompson, R.N., et al. (2020). Estimating the time-varying reproduction number of SARS-CoV-2 using national and subnational case counts. *Wellcome Open Research*, 5, 112.

[10] Birrell, P., Blake, J., van Leeuwen, E., et al. (2021). Real-time nowcasting and forecasting of COVID-19 dynamics in England. *Statistical Science*, 36(4), 509–531.

[11] Flaxman, S., Mishra, S., Gandy, A., et al. (2020). Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe. *Nature*, 584, 257–261.

[12] Brauner, J.M., Mindermann, S., Sharma, M., et al. (2021). Inferring the effectiveness of government interventions against COVID-19. *Science*, 371, eabd9338.

[13] Grünwald, P., van Ommen, T. (2017). Inconsistency of Bayesian inference for misspecified linear models, and a proposal for resampling-based model checking. *Bayesian Analysis*, 12(4), 1069–1103.

[14] Sherratt, K., Gruson, H., Grah, R., et al. (2023). Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. *eLife*, 12, e81916.

[15] Bolzoni, L., Bonacini, E., Soresina, C., Groppi, M. (2017). Time-optimal control strategies in SIR epidemic models. *Mathematical Biosciences*, 292, 86–96.

[16] Bliman, P.-A., D'Avino, G. (2021). How best can finite-time social distancing reduce epidemic final size? *Journal of Theoretical Biology*, 511, 110557.

[17] Rawlings, J.B., Mayne, D.Q. (2009). *Model Predictive Control: Theory and Design*. Nob Hill Publishing, Madison, WI.

[18] Schwarm, A.T., Nikolaou, M. (1999). Chance-constrained model predictive control. *AIChE Journal*, 45(8), 1743–1752.

[19] Betts, J.T. (2010). *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming* (2nd ed.). SIAM, Philadelphia.

[20] Zhu, C., Byrd, R.H., Lu, P., Nocedal, J. (1997). Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software*, 23(4), 550–560.